Supplementary Information

Harnessing Plastic Deformation for Continuous Passive Stability in Hybrid Metal–Elastomer Structures

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Movie S4. Rapid Grasping.

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Movie S10. Quadruped Robot Locomotion on a Tube.

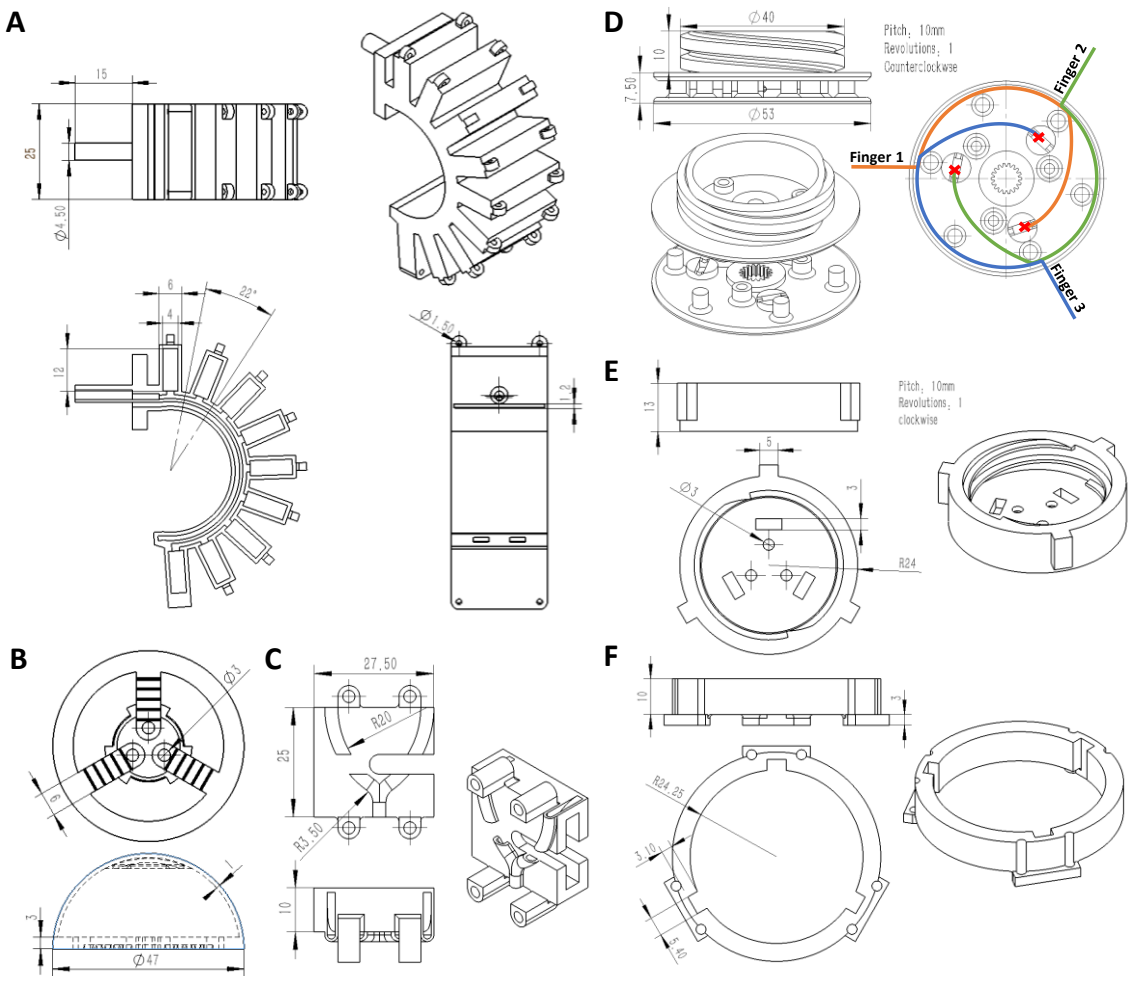


Fig. S1. Schematic representation of the component dimensions. (A) Soft finger; **(B)** Paw pad; **(C)** Soft finger fixture; **(D)** Screw; **(E)** Flanged nut; **(F)** Linear guide. Due to the thermoplastic effect of TPU, the initial angle of the soft finger 3D printed will be smaller than the set pre-stretching angle.

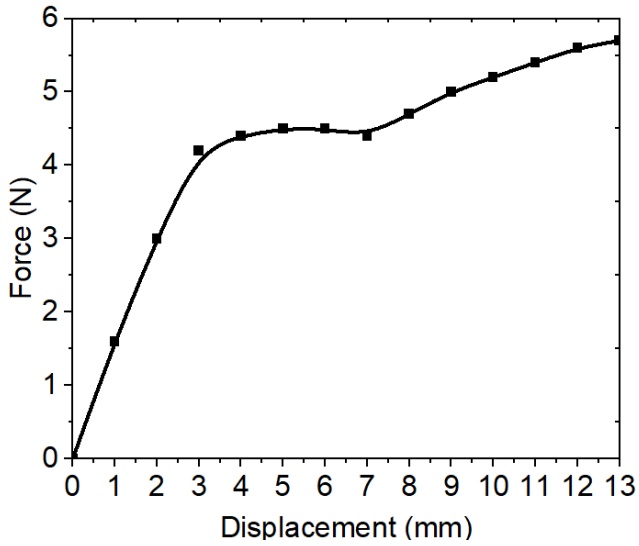


Fig. S2. Force-displacement characteristics of the paw pad. The sensor exhibits high sensitivity to external forces at the initial stage of displacement, allowing for enhanced detection of object contact. When the displacement exceeds approximately 4 mm, the sensitivity decreases, indicating a transition to a stable support phase that provides a consistent supporting force of around 4.5 N for the grasped object.

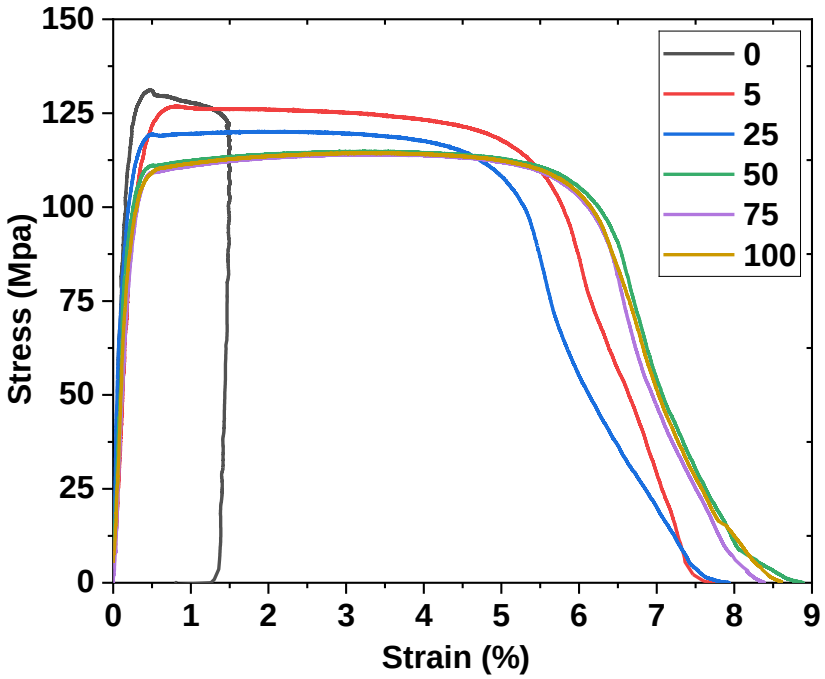


Fig. S3. The result of the tensile test with different numbers of bending cycles.

We conducted tensile tests on aluminum samples subjected to different numbers of bending cycles (0, 5, 25, 50, 75, and 100). To investigate the effect of repeated bending on their tensile behavior, the samples were subjected to uniform 90° bending for the specified number of cycles, and their mechanical responses were subsequently evaluated. As shown in the figure, after the first bending cycle, the fracture strain increased significantly, whereas the ultimate tensile strength decreased. With an increasing number of plastic deformation cycles, the stress-strain relationship gradually stabilized, achieves cyclic saturation (a shakedown state).

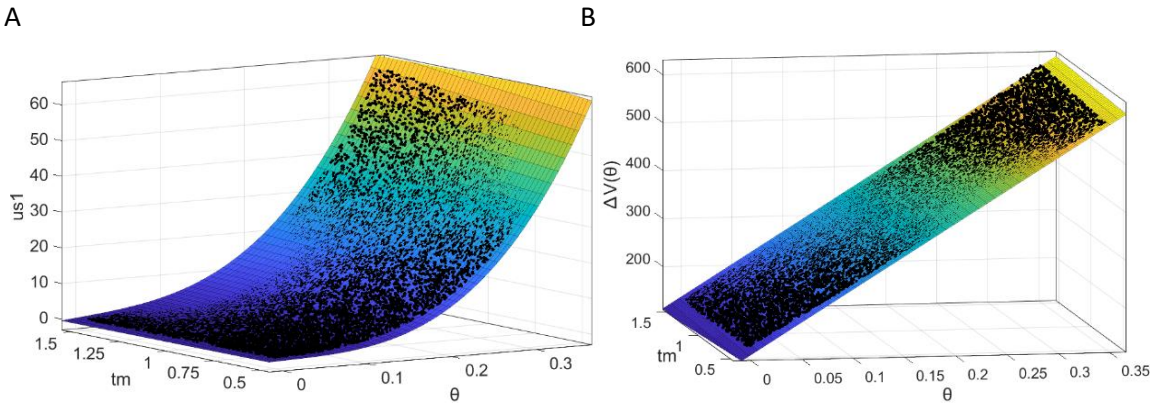


Fig. S4. Surface fitting visualization. **(A)** The fitted result of the integrand corresponding to the integral function $U_{s1}(\theta)$, R-square: 0.9999, RMSE:0.1572; **(B)** The fitted result of the integrand corresponding to the integral function $\Delta V(\theta)$, R-square: 0.9999, RMSE: 0.8883.

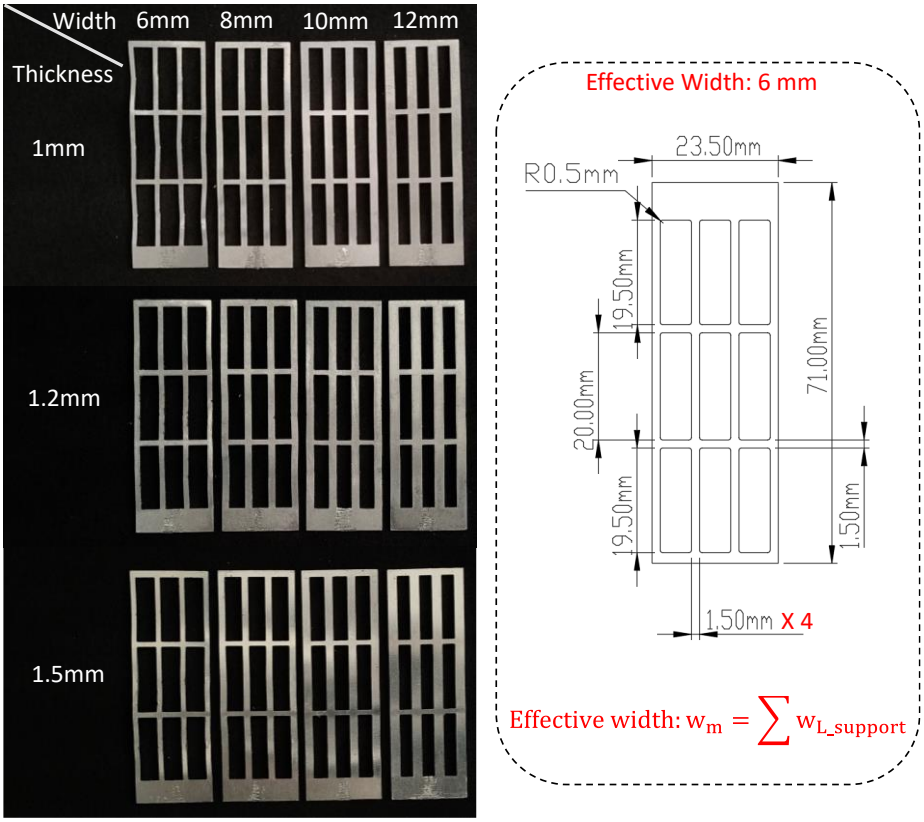


Fig. S5. Metal layers with different thickness and width after the experiments.



Fig. S6. The sample and machine of the tensile test. (A) Dimensions of the dumbbell-shaped tensile sample. **(B)** Instron 5982 mechanical testing machine. **(C)** Fractured tensile samples after the uniaxial tensile test.

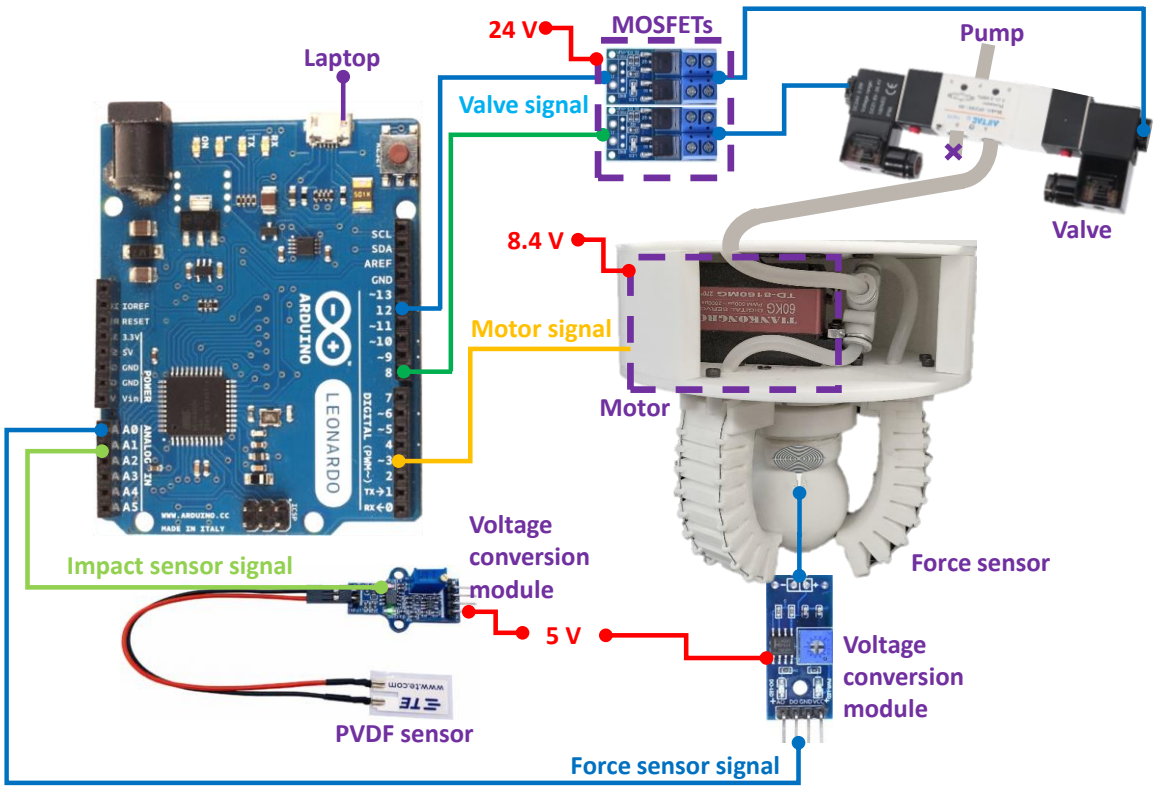


Fig. S7. The control system scheme.

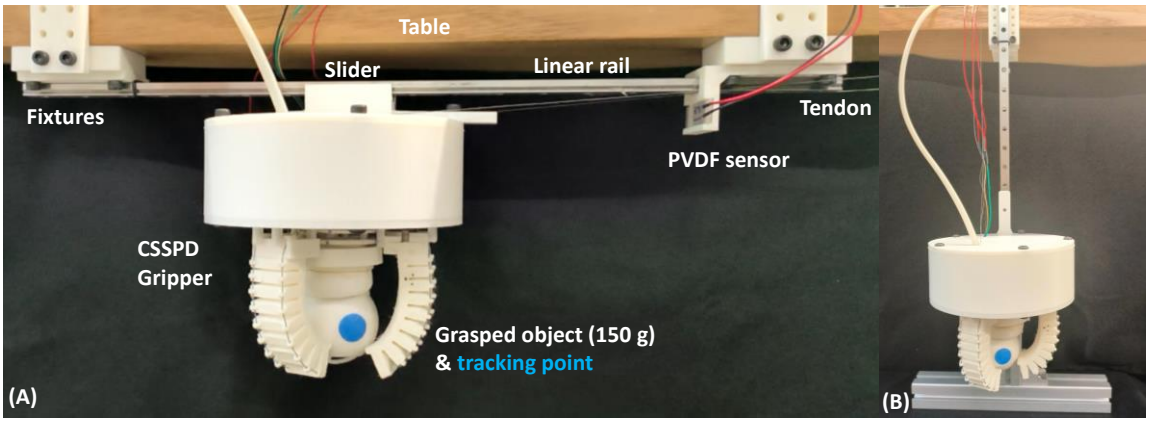


Fig. S8. Experimental setup. (A) Horizontal stability test platform. **(B)** Vertical stability test platform**.** The impact force F and impact duration Δt can be obtained using the PVDF sensor. Due to the high sensitivity of the PVDF sensor, the measured impact time is very short, averaging around 10 ms. Therefore, the measured force can be regarded as the actual instantaneous force during the impact, and the resulting acceleration can be calculated using $a = F/m$.

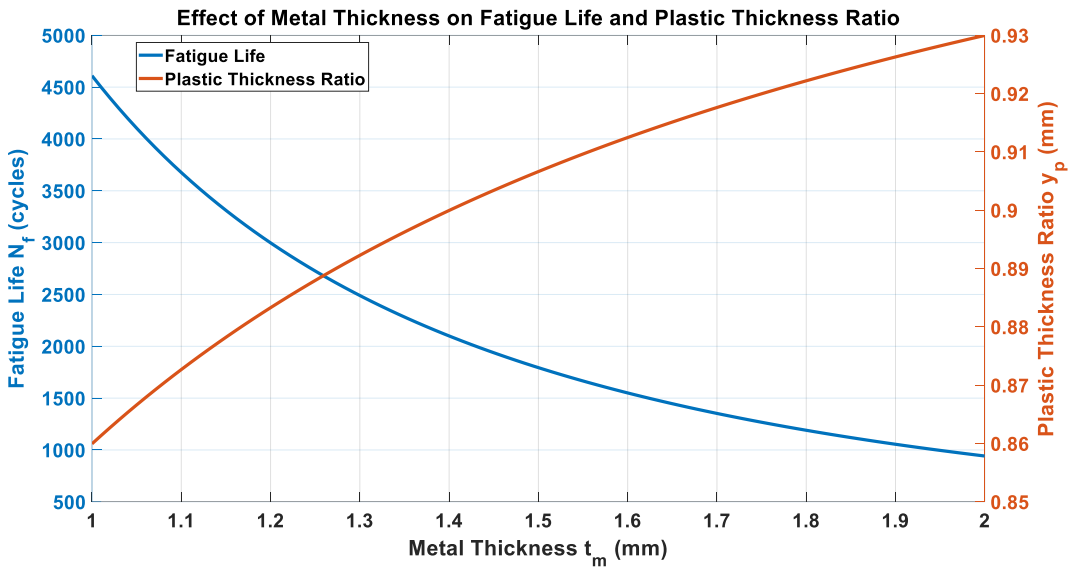


Fig. S9. Effect of metal-layer thickness on fatigue life and plastic deformation behavior.

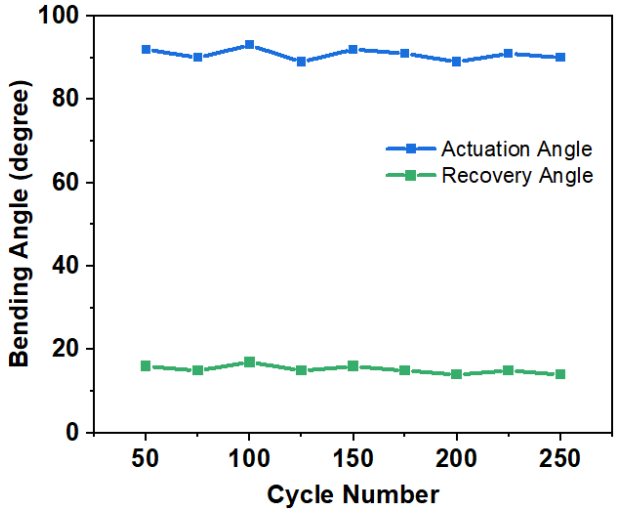


Fig. S10. Cyclic stability of the actuation and recovery angles over 250 actuation cycles.

Table S1. Key slicing parameters for 3D printed TPU material.

Layer height	Infill density	Ironing type	Ironing flow	Wall loops	Infill overlap	First layer speed	Other layers speed	Bridge & Gap infill speed
0.1 mm	100 %	All solid	30%	15	30%	30 mm/s	30 mm/s	30 mm/s

Table S2. Chemical composition of 1060 ALU.

Si	Fe	Cu	Mn	Mg	V	Zn	Ti	Al
0.05%	0.03%	0.009%	0.002%	0.002%	0.005%	0.004%	0.002%	Balance

Table S3. Structural and material parameters of the gripper.

Para.	Value	Para.	Value
Finger total length	90 mm	Tendon diameter	0.4 mm
Finger total width	25.5 mm	Motor torque	60 kg·cm
Number of segments	9	Finger mass	0.012 kg
Total prestretching angle	198°	Metal mass	0.002 kg
Metal layer length	71 mm	Gripper total mass	0.66 kg
Metal layer effective width	6 mm	TPE density	1140 kg/m^3
Metal layer thickness	1.2 mm	Metal density	2700 kg/m^3
Paw pad diameter	47 mm	TPE 1st Yeoh material para.	3.2 MPa
Servo horn diameter	42 mm	Metal elastic modulus	60 GPa
Tendon routing distance	18 mm	Metal yield stress	0.5 GPa

Table S4. Comparison of the proposed gripper with other stable grippers.

	Energy-free holding	Self-adaptability after grasping	High Speed	Continuous deformation	Energy-free Stiffness
This work	√	√	√	√	√
Bistable/ Multi-stable	√	√	√	×	√
Granular jamming	×	×	√	√	√
Layer jamming (Friction)	√	√	×	√	√
Phase change	√	×	×	√	√
SMA//SMP	×	√	×	√	×
Ratchet	√	×	√	×	√

Supplementary Text S1. The optimization of metal layer.

The design of the kirigami metal layer was formulated as a multi-objective optimization problem, where the thickness and width of the metal layer were selected as the primary design variables:

$$\mathrm{X} = [t_m, W_m] \tag{S1}$$

Where t_m and W_m represent the thickness and total width of the metal layer, respectively.

Three competing objectives were considered:

$$\begin{cases} f_1 = N_f \\ f_2 = \gamma_{ratio} \\ f_3 = M_{\mathrm{t,max}} \end{cases} \tag{S2}$$

where N_f represents the fatigue life of the metal layer, γ_{Ratio} denotes the plastic thickness ratio region to the total metal thickness, and $M_{\mathrm{t,max}}$ represents the maximum tendon-driven torque required for actuation

The plastic deformation ratio is defined as:

$$\gamma_{Ratio} = \frac{t_m - 2y_{\mathrm{p}}}{t_m} \tag{S4}$$

where y_{p} is plastic region which can be expressed as (same as Eq. 10):

$$y_{\mathrm{p}} = \frac{h\sigma_{\mathrm{yield}}}{E\theta} = \frac{hE\varepsilon_{\mathrm{yield}}}{E\theta} \tag{S5}$$

The optimization objectives involve inherent trade-offs. Increasing the metal thickness decreases fatigue life by increasing the stress on the outter surface of the metal layer. It simultaneously increase the relative plastic deformation ratio and increases the bending resistance, resulting in a higher actuation torque. Therefore, the optimal design requires balancing shape retention, mechanical durability, and actuation requirements.

The optimization was performed under the following constraints:

$$\begin{cases} 1 \le t_m \le 2\ mm \\ 6 \le W_m \le 24\ mm \\ 0 \le \gamma_{Ratio} \le 1 \\ \theta_{max} = \pi/2 \end{cases} \tag{S6}$$

where the thickness and width ranges were determined according to fabrication limitations and the required bending performance of the soft finger.

Considering that the fatigue life and plastic deformation ratio are primarily governed by t_m, while W_m mainly affects the required driving torque, once t_m is determined, W_m can be reasonably selected to match the chosen servo and servo disc, the thickness was first optimized within the feasible range. The influence of the metal thickness on the optimization objectives was first investigated, as shown in Fig. S9. A thickness of metal $t_m = 1.2\ mm$ was selected as a compromise between sufficient fatigue resistance and adequate plastic deformation capacity.

Within the operational workspace of 0 to $\pi/2$, W_m was first set to its minimum value of 6 mm. The maximum tendon-driven torque is given by:

$$M_{\mathrm{t,max}} = M_{\mathrm{m}}\left(\frac{\pi}{2} - \theta_0\right) - M_{\mathrm{PL}}(\theta_0) = 694.5\ \mathrm{N \cdot mm} \tag{S7}$$

Given that the distance from the tendon routing point to the constant layer is 18 mm, the required tendon force is calculated to be 38.6 N. Since the system simultaneously drives three fingers, and considering the weight and size constraints of the servo motor, we selected the TD-8160MG servo, which provides a torque of 60kg▪cm. Based on this, the maximum required servo horn radius was determined to be 50 mm. To ensure safety, we ultimately opted for a 3D-printed servo horn with a radius of 21 mm, which meets the design requirements. Therefore W_m was ultimately set to 6 mm.

Supplementary Text S2. The fatigue life of the metal layer.

Because reliable fatigue testing of the embedded metal layer requires long-term cyclic loading under well-controlled and repeatable deformation conditions, its fatigue performance is evaluated analytically using the proposed mathematical model. Unlike axial tension, where deformation is distributed across the material, in bending, the maximum strain is localized only on the outer surface, and the strain magnitude remains relatively small, classifying it as a low-cycle fatigue (LCF) process. The theory of critical distances (TCD) [1], combined with the Coffin-Manson equation [2], provides a more accurate evaluation of the fatigue life of the metal sheet. To avoid overestimating the fatigue life, a truncated integration approach is adopted, in which the effective strain ε_{eff} is averaged only over the smaller of the physically available thickness $t_m/2$ and the critical distance d. Then the strain distribution ε_{zz} through the thickness is described by Eq. (S8). By calculating the equivalent strain ε_{eff} (Eq. S9), the critical distance d (Eq. S10), and applying the Coffin-Manson equation (Eq. S11), the fatigue life N_f can be determined.

For our analysis, the bending angle θ_{max} is set to the maximum workspace of π/2, with the metal layer length $h = 60$ mm. The fracture toughness $K_{\mathrm{IC}} = 30\mathrm{MPa}$. The thickness of the metal layer $t_m = 1.2$ mm. After repeated bending beyond the elastic limit, leading to plastic deformation and work-hardening (akin to cold working, see Fig. S3), yield stress $\sigma_y = 110$ MPa, elastic modulus $E = 60$ GPa. And fatigue strength coefficient $\sigma`_{\mathrm{f}} = 200\mathrm{Mpa}$, fatigue strength exponent $b = -0.1$, fatigue ductility coefficient $\varepsilon`_{\mathrm{f}} = 0.5$, fatigue ductility exponent $c = -0.5$, material constant $\alpha = 0.2$. The parameters selected based on reported ranges for commercially pure aluminum fatigue studies [3]. Using these parameters, the calculated conservative fatigue life is $N_{\mathrm{f}} \approx 3000$. Selecting materials with better fatigue properties and increasing the metal layer thickness can further enhance the fatigue life.

$$\varepsilon_{\mathrm{zz}} = \frac{\theta_{max} y}{h} \tag{S8}$$

$$\varepsilon_{\mathrm{eff}} = \frac{1}{\min(\mathrm{d}, t_m/2)} \int_0^{\min(\mathrm{d}, t_m/2)} \varepsilon_{\mathrm{zz}}\, \mathrm{d}y \tag{S9}$$

$$d = \alpha\left(K_{\mathrm{IC}}/E\varepsilon_{\mathrm{y}}\right)^2 \tag{S10}$$

$$\varepsilon_{\mathrm{eff}} = \frac{\sigma`_{\mathrm{f}}}{E}(2N_{\mathrm{f}})^{\mathrm{b}} + \varepsilon`_{\mathrm{f}}(2N_{\mathrm{f}})^{\mathrm{c}} \tag{S11}$$

Future work could further improve system performance and fatigue life through the selection of more suitable metallic materials. Another alternative approach is to integrate reconfigurable metallic materials, such as low melting point alloys, which can erase deformation history through thermally induced melting and resolidification, thereby enabling reversible structural reconfiguration and improving the durability and adaptability of the system.

Supplementary Text S3. Mechanical stability criterion.

To clarify the mechanical meaning of continuous stability, we define a passively stable configuration as a post-unloading equilibrium state in which the net bending moment is zero and a small perturbation generates a restoring moment. Accordingly, the equilibrium condition can be expressed as

$$M_{res}(\theta_f) = 0 \tag{S12}$$

where M_{res} denotes the resultant bending moment after unloading and θ_f is the final bending angle. The local stability of this equilibrium is determined by the corresponding tangent stiffness,

$$K_{\text{eff}} = \left.\frac{\partial M_{res}}{\partial \theta}\right|_{\theta=\theta_f} > 0 \tag{S13}$$

A positive K_{eff} indicates that a small perturbation from θ_f generates a restoring moment toward the equilibrium configuration, and therefore the configuration is locally stable.

Importantly, plastic deformation of the embedded metal layer does not itself provide a restoring force. Instead, it permanently changes the residual geometry of the metal layer and thereby shifts the equilibrium configuration. After unloading, the elastic recovery of the surrounding soft material and the bending response of the plastically deformed metal layer counteract each other to establish a new equilibrium. This is consistent with the bending-stiffness formulation given in Eq. (36), where the effective stiffness is determined by the combined contributions of the pre-stretching layer and the elastic metal component. Therefore, a stable post-unloading configuration satisfies both the equilibrium condition $M_{res} = 0$, and the local stability condition $K_{\text{eff}} > 0$.

In this work, continuous stability refers to the existence of such passively stable equilibrium configurations over a continuous range of bending angles,

$$\theta_f \in [\theta_{min}, \theta_{max}] \tag{S14}$$

rather than at only a finite number of discrete locking positions. Unlike bistable or multistable mechanisms, whose stable configurations are associated with discrete energy minima, the CSSPD mechanism continuously modifies its residual configuration through plastic deformation. Consequently, different input deformation levels can produce different passively retained equilibrium configurations within the allowable deformation range, without continuous energy input or predefined mechanical locking positions.

Supplementary Movies

Movie S1. Behavior of a Soft Finger. A single soft finger with an embedded metal sheet is manually bent to arbitrary angles in the unpressurized state. It is then actuated at different pressures while fixed to a platform. The video demonstrates that the CSSPD maintains continuous and stable deformation.

Movie S2. Adaptive Grasping. The gripper, embedded with metal sheets, grasps a 150 g spherical object under pressurization. After deflation, the biomimetic paw pad adapts to the object and provides sustained upward support. This compensates for angular recovery caused by plastic deformation of the metal sheets, allowing the gripper to securely hold the object even after deflation and to resist external disturbances.

Movie S3. Paw Pad Structure Sensing. A force sensor embedded in the biomimetic paw pad accurately detects external forces during pad deformation, validating its sensing capability.

Movie S4. Rapid Grasping. With the paw pad structure, the gripper can detect object impact via its force sensor. When the impact force exceeds a predefined threshold, the system automatically opens the pneumatic valve, enabling rapid and autonomous grasping.

Movie S5. Static Stability. The gripper is fixed to a platform and used to grasp a 150 g object under three different conditions: (i) no metal layer with 0.3 MPa pressure, (ii) metal layer without actuated pressure, and (iii) metal layer with 0.3 MPa. A force gauge is used to pull the object slowly. The results show that the stability provided by metal sheets alone is comparable to that under 0.3 MPa pressure, and that combining metal sheets with pressurization significantly enhances static stability.

Movie S6. Vertical Dynamic Stability. A vertical rail is mounted on the platform. The gripper, fixed to a slider and grasping a 150 g object, is released from the same height to free-fall and collide with a sensor. Pulsed acceleration is calculated based on impact force measurements. The video shows the comparison at a pulsed acceleration of 400 m/s^2. The object's motion trajectory is extracted using MATLAB.

Movie S7. Horizontal Dynamic Stability. A horizontal rail is mounted on the platform to conduct lateral dynamic stability tests. The experiment evaluates the gripper's performance during horizontal impacts while grasping.

Movie S8. Grasping Diversity Experiment. The gripper successively grasps a variety of objects with different sizes, weights, and shapes, demonstrating its adaptability and versatility.

Movie S9. Energy-Free Stable Holding Capability. The gripper is used as a mobile platform. It grasps both an inclined branch and a vertical rod.

Movie S10. Quadruped Robot Locomotion on a Tube. The CSSPD fingers are arranged in a quadruped robot configuration, enabling locomotion along a cylindrical surface. A solar panel mounted on the robot can be deployed using shape memory alloy (SMA) actuation.

Referencces

[1] D. Taylor, The Theory of Critical Distances. Elsevier Science Ltd, 2007.

[2] K. J. J. o. m. Smith, "A stress-strain function for the fatigue of metals," vol. 5, pp. 767-778, 1970.

[3] H. J. A. M. Mughrabi and Processes, "ASM Handbook, Volume 19, Fatigue and Fracture," Advanced Materials & Processes, vol. 153, no. 1, p. 76, 1998.